

Pencil-E: Crafting Functional Electronics Using Pencils and Paper

Yiyang Li
Zhejiang University
Hangzhou, China
yiyangli@zju.edu.cn

Jiazheng Yu
Zhejiang University
Hangzhou, China
3190105246@zju.edu.cn

Tianze Guo
Zhejiang University
Hangzhou, China
3200102554@zju.edu.cn

Hao Dong
Rutgers University
New Jersey, United States
hd322@scarletmail.rutgers.edu

Jiajun He
Zhejiang University
Hangzhou, China
jiajunhe@zju.edu.cn

Ting Zhang
Zhejiang University
Hangzhou, China
zhangting5@zju.edu.cn

Guanyun Wang*
Zhejiang University
Hangzhou, China
guanyun@zju.edu.cn

ABSTRACT

We present Pencil-E, a low-cost, ubiquitous, and accessible sensing and actuating method using pencils and paper for personal fabrication. The basic architectures of this method are the principal electrical features of the pencil as a conductive material and the properties of the paper as a carrier of pencil traces. By combining paper craft methods, including single-layer, multi-layer, and constructive structures, we developed resistive and capacitive sensors with haptic feedback and allowed customization. This paper describes the design guidelines, sensing mechanics, primitive constructions, and the fabrication process. Multiple pre-designed elements and application scenarios are demonstrated, including tangible user interfaces, wearable devices, and interactive art crafts.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods**; **Human computer interaction (HCI)**.

KEYWORDS

pencil, paper interface, flexible electronics, personal fabrication, rapid prototyping

ACM Reference Format:

Yiyang Li, Tianze Guo, Jiajun He, Jiazheng Yu, Hao Dong, Ting Zhang, and Guanyun Wang. 2023. Pencil-E: Crafting Functional Electronics Using Pencils and Paper. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585759>

*Corresponding author

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585759>

1 INTRODUCTION

Flexible electronics created by personal fabrication have become an essential part of Do-It-Yourself interactive circuits, as they are free-form, deformable, and apply to multiple materials. They are used in technical artworks [1], wearable devices [6, 15, 18, 26], customized interfaces [3–5, 23], and personal-fabricated electronic items [14, 17]. The common types of personally fabricated electrical flexible circuits can categorize into three types. 3D-printing electrical circuits use 3D printers to embed conductive filaments for accurate, complex parts. Textile-based circuits use conductive yarn knitted in patterns for wearable, washable sensing interfaces. Paper-based circuits use copper foil tape, and silver ink for interactive papercrafts.

Skills in certain areas, costly equipment, and some embarrassing materials, either of the three conditions are required for the above method. However, our work presents a new way of thinking about how to meet the demand for flexible circuits with the lowest effort and cost. We focused on a handmade process and design-driven reflection on basic materials to break the black box of industrialized sensors and lower the technical and resource barrier of creating a DIY sensor. With this method, we hoped to enable a broader community to easily create their own sensing components based on their desired structure or interactive mode ubiquitously.

The Pencil-E method presents an approach to designing and fabricating flexible, low-cost, and tangible sensors with customized interactive methods using pencil and paper. Pencil-E focused on two sets of features in both materials. Pencil, a mixture of graphite and clay, is seen as a conductive material and as a tool for artistic creation [19]. Paper, with diverse properties in roughness and surface strength, is a great substrate to pill down the graphite chunks which deposit as pencil traces and is used as a functional structure provider as well as a tangible feedback generator.

The mechanics were based on the resistive and capacitive features of the pencil and the haptic primitives of kirigami and origami structures. We leveraged three types of pencil-paper combinations: single-layer, multi-layer, and constructive-layer. Users can create flexible circuits based on the fundamental constructions provided using the most ubiquitous materials to draw, cut, and fold. This

will enlarge the application space of electronics to become more diverse and accessible. The workflow of Pencil-E includes sketching the circuit design on paper with 8B to 16B pencils, processing the design pattern into a valid structure, and then connecting it to the leading circuit with alligator clips or copper foils. We provide tools to generate vector toolpath and design guidelines to facilitate users in designing and fabricating unique Pencil-E structures. Finally, we explored the design possibility of functional paper craft using the method.

The main contributions of this work are as follows:

- A paper sensing technology using low-cost material to fabricate tangible interactive electronics, introducing a new approach to enrich the library of tangible HCI components.
- Extend the design field by introducing the paper craft to create a single-layered, multi-layered, and three-dimensional circuit.
- Applications using the Pencil-E method that demonstrates the design space in both functional and artistic ways.

2 RELATED WORKS

2.1 Paper Electronics

The paper has immense potential to replace standard electronic interfaces for HCI researchers. It is inexpensive, lightweight, easy to use, and readily available. Along with the development of patch electronics and the growth of paper crafting art, paper enables more and more users to create their own devices in both practical and artistic ways, making paper electronics a burgeoning field. Some use paper as a substrate, adding additional functional units onto it [11, 20]. Some brought artistic sense into paper with the help of electronics [8]. Others fully explore the boundaries of paper structures by using origami, kirigami structures to create haptic sensors [1, 17, 27] or piling up papers to get three-dimensional electronics [16]. Previous works mainly focused on paper as a substrate for electronics, but we hope to explore the potential of paper structure as a means to support fundamental electronic mechanics.

2.2 Fabrication of Conductive Materials

Traditionally, technology enthusiasts have used standard materials such as DuPont lines on breadboards or soldered components onto protoboards or printed circuit boards to build electronics. However, with advances in low-cost personal fabrication, more ubiquitous materials are being used to create functional circuits. Researchers have thoroughly studied the physical and chemical mechanics of graphite pencils as a conductive material [9, 13, 21]. Sheet metals such as gold leaves and copper foil are also used to create conductive paths [10]. Artistic expression and circuit design can be combined using conductive ink which forms the circuitry and serves as part of the illustration [7, 19]. However, existing methods still heavily rely on uncommon materials or precise detection equipment. Our work followed this path of creative circuitry and focused on exploring a more accessible and widely-used conductive material and mechanics: sensing the graphite pencil.

2.3 Multi-layer Electronics

In industrial manufacturing, multi-layer electronics offer many advantages, such as high density [25], high performance [2], and high scalability [22]. This technology is widely used in HCI and has enabled the creation of sensing textiles through machine knitting [15], shape-changing objects with foldable printed electronics [17], thermoformed circuit boards [5, 24], and layered circuit manufacturing [23]. Our work explored the use of diverse papers in constructing multi-layer circuits, greatly expanding the use of multi-layer electronics and enriching the interaction modes.

3 PENCIL-E METHOD

In this section, we will cover the fabrication principle of using pencil and paper as the fundamental material, as well as the sensing mechanics and standard structures of the Pencil-E method.

3.1 Pencil-E Material

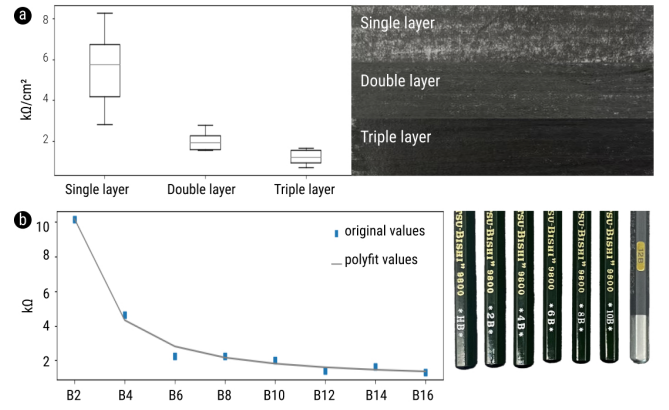


Figure 1: (a) Resistance changes according to pencil trace thickness; (b) resistance changes according to pencil grade.

3.1.1 Pencil Trace Thickness. The number of layers the pencil applied over the same area affects the resistance of a pencil-covered area. To evaluate, we divided samples into 3 groups (single, double, and triple layers) and measured resistance per unit area using a multimeter. The pencil traces are applied along the length of 40mm by 20mm, 60mm by 20mm, 80mm by 20mm, 40mm by 30mm, 60mm by 30mm, and 80mm by 30mm handcraft paper strips. Results show that resistivity is lower and more stable with double and triple layers (Figure 1-a). A single-layered coating will cause inconsistent deposits leading to partial disconnections. We recommend hatching twice for a more consistent pencil-paper connection.

3.1.2 Pencil Grade. The blackness grade (B grade) of the pencil directly influences the conductivity of a pencil trace. To evaluate, the pencil traces are applied along the length of 40mm by 20mm handcraft paper strips, double-layered. Our study found that resistance decreases as the B grade of pencils increase on handcraft paper strips ($y = 17.093x^{-0.9508} + 1.249$, $R_2 = 0.9469$), indicating that higher blackness pencil marks have better conductivity (Figure 1-b). We recommend using a higher blackness pencil to reach the resistance value of the target efficiency.

3.1.3 The Direction of Pencil Trace. The direction in which pencil lines are drawn can also affect conductivity. The fundamental mechanism is the same as in the single-layer rotating structure. Some design tips to consider include: (1) drawing lines along the flow of electric current can lead to the best capacity results; (2) Cross-hatching techniques can reduce the anisotropy (uneven conductivity in different angles) of a pencil-filled area.

Available Paper				Unavailable Paper
Paper Surface	Sulfate Paper	Corrugated Paper	Handcraft Paper	Copperplate Paper
Resistance	1.26k Ω	1.59k Ω	1.45k Ω	160k Ω
Resistance After Folded Once	2.97k Ω	25.2k Ω	160k Ω	1760k Ω
Paper Surface	Cardboard Paper	Scrub Tape		Heavy cardboard
Resistance	2.6k Ω	6.95k Ω		16.42k Ω
Resistance After Folded Once	-	38.9k Ω		-

Figure 2: The resistance of different papers and the resistance performance after bending.

3.1.4 Paper as a pencil substrate. Rough surfaces perform better for conductivity and a surface roughness of 1-5 μm is required to attach graphite chunks [12]. To facilitate personal fabrication, we studied the relationship between paper surface roughness and the conductivity of pencil traces. We used a 16B pencil and applied it twice along the 20mm by 40mm paper strips. Results show that sulfate paper, corrugated paper, and handcraft paper are recommended for Pencil-E constructions (Figure 2). Additionally, it is found that the paper's thickness and fiber types affect the resistance while being bent. Sulfate paper performs best in maintaining conductivity, and corrugated paper is recommended as a stable material with haptic feedback.

3.2 Sensing Mechanics

The sensing mechanic can be divided into resistive and capacitive sensing. Conductive pencil traces can be used to create resistance sensors that can detect changes in (a) the density of graphite chunks (Figure 3-a1), (b) the length of connected pencil traces (Figure 3-a2), or (c) the connection state (Figure 3-a3). We measured resistance using a voltage divider circuit composed of our resistance sensor, an Arduino Uno board, and a Keyes's voltage sensor.

Pencil tracks can retain charge and are used to create capacitive sensing structures by drawing intersecting doodles on different overlapping sides. Two types of capacitive sensors are designed based on changes in (a) inter-plate distance (Figure 3-b1) or (b) overlapped plate area (Figure 3-b2), which is directly measured using an Arduino Uno board.

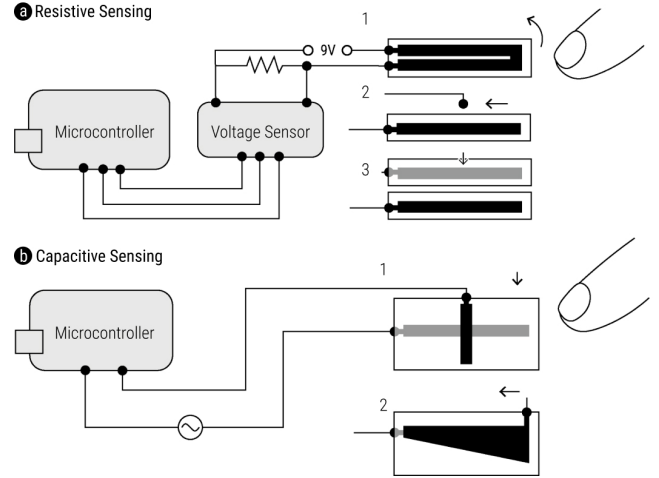


Figure 3: (a) The sensing mechanics of resistive sensors; (b) the sensing mechanics of capacitive sensors.

3.3 Primitive Structures

We bring variation to sensors by combining paper structures with sensing mechanics. We focused on the diversity of both appearances and types of interactions.

3.3.1 Single-layer Structure. Single-layer type is a mixture of a single layer of pencil track on a paper scrap, it can be interacted with by folding and rotating (Figure 4-a).

Folding type changes graphite chunk density in pencil trace. Curving back and forth causes graphite chunks to become tense and loose, increasing or decreasing resistance (Figure 4-a1). Sharp and curved folds have the same effect, but sharp folds are more prone to wear and tear and may break the circuit after several repetitions (Figure 4-a2).

The rotating sensor is made with pencil tracks drawn in the same direction (Figure 4-a3). Resistance changes occur as the angle between the conductive trace and current trace change, due to the density of the graphite chunks being higher resistance within a pencil track than between them.

3.3.2 Multi-layer Structure. Multiple layers of paper and pencil are assembled to create more complicated, yet more productive components, such as capacitive features (Figure 4-b).

The pattern type creates a capacitive touch sensor using only pencil doodles with minimal restrictions. It consists of a paper scrap with overlapping pencil patterns on both sides. The sensor in Figure 4-b1 detects changes in the distance between the plates caused by pressure. The higher the touch pressure, the greater the deformation of the paper thickness. The sensor in Figure 4-b2 is a touch slider that detects changes in the deformed area.

The kirigami type uses kirigami structures to function as clips that hold the paper in place and allow rotation. We utilize this feature to create (a) a capacitive sensor that changes the overlapping pencil area (Figure 4-b3) and (b) a resistive sensor whose length of conductive material changes with rotation (Figure 4-b4).

The stack-up type piles multiple layers of paper together to create multi-layer circuits that save space and materials and introduce new

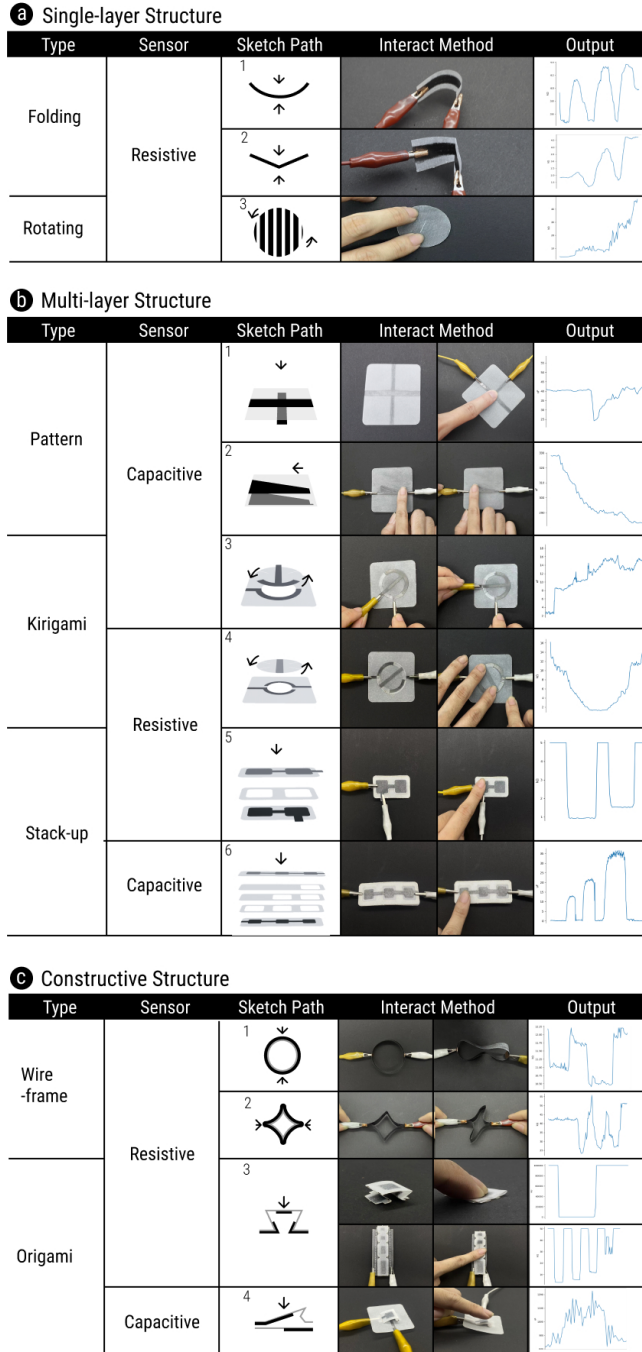


Figure 4: Sensing structures.

functions and aesthetic potentials. The sensor in Figure 4-b5 is a resistive sensor that utilizes the property of pencil tracks requiring pressure to connect to a conductive material. The capacitive sensor in Figure 4-b6 uses cavities of different depths to create a change in capacity value when pressed.

3.3.3 Constructive Structure. By combining paper into 3D structures, we designed constructive structures (Figure 4-c). The wire-frame type is a combination of multiple single-layer paper strips that can be activated by one-dimensional force and stacked to create aesthetic crafts (Figure 4-c1,c2). The origami type use paper folding to create low-cost and easy-to-make 3D electronic structures with haptic feedback (Figure 4-c3,c4). The press-and-release action creates binary signals in a basic unit, which can be duplicated to create a slider, as shown in Figure 4-c3. Pressing different positions of the slider unit changes the conductive path, resulting in a detectable change in resistance.

4 DESIGN AND FABRICATION

4.1 Design Interface

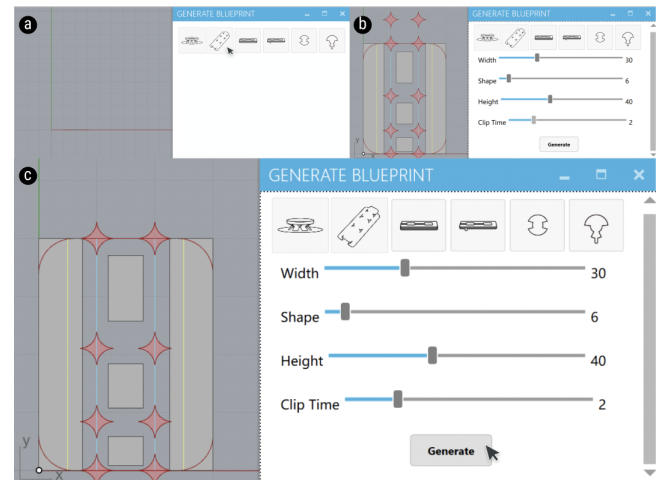


Figure 5: Design walkthrough: (a) choose the form; (b) adjust the elements; (c) generate printing tool paths.

Paper folds, cuts, and closed outlines of pencil traces can describe a Pencil-E pattern. Colored lines can be used as hints for handcrafting or as tool paths for laser-cutting machines. To simplify the fabrication process, we designed a specialized tool for creating Pencil-E elements using Rhinoceros 6 and Grasshopper platforms. This tool includes six structures to choose from and several adjustable parameters to customize the pattern.

4.1.1 Choose the form. We offer six fundamental types of sensors, including the resistive button/slider, resistive knob, resistive touchpad, capacitive touchpad, resistive radio button, and capacitive radio button. Users can select a specific form by clicking the icon button, which generates an outline of a default element (Figure 5-a).

4.1.2 Adjust the elements. Users can adjust the length, width, and sensor precision using the GUI slider. For instance, the segments of the resistive button or slider, the number of keys of the resistive knob, and the touchable areas of the resistive and capacitive touchpad (Figure 5-b).

4.1.3 Generating printing toolpaths. Lastly, the software can automatically generate the toolpath for printing and laser-cutting. Additionally, a base plate for clipping the elements will be generated, and users can freely adjust the position to fit their personal objectives (Figure 5-c).

4.2 Fabrication Process

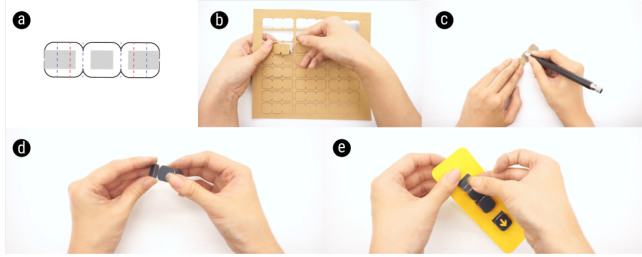


Figure 6: The fabrication process of Pencil-E.

To make it easy for personal use, we have outlined the fabrication process as follows: (a) Use design software to generate the vector graphic of components (Figure 6-a), and print or draw it on suitable paper; (b) cut out the shape following the black guidelines (Figure 6-b); (c) follow the gray lines and use a pencil to fill in the closed area (Figure 6-c); (d) fold paper following color lines if needed: blue indicates valley folds, and red indicates mountain folds (Figure 6-d); (e) clip the structure into a base plate and connect it to a microcontroller board for signal reception (Figure 6-e).

5 EVALUATION

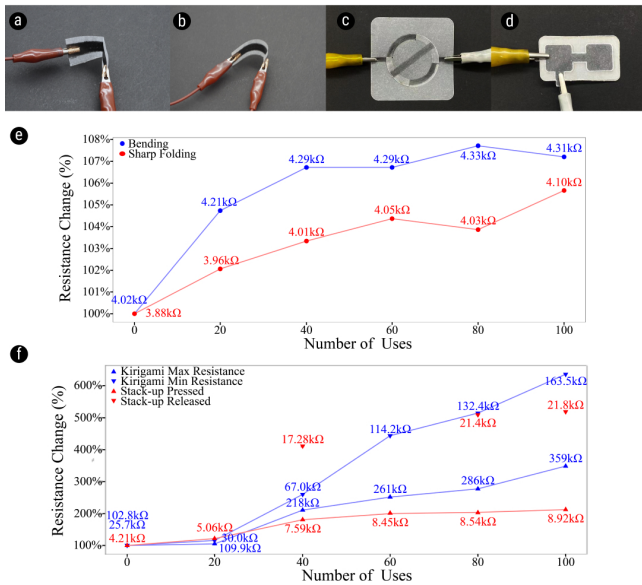


Figure 7: Structures to evaluate and recorded resistance change.

The graphite layer is fragile and fast-wearing, making pencil-paper structures gradually lose sensing ability. We evaluated the durability of four basic resistive structures by measuring resistance change after repeated use. Capacitive methods were not tested as capacitance is not affected by the abrasion of the graphite layer. The structures we tested are the single-layer folding type (Figure 7-a,b), the kirigami type (Figure 7-c) as well as the stack-up type (Figure 7-d) in multi-layer structures. We measured resistance every 20 uses, from 0 to 100 uses. One use was defined as: bending inward and outward to 90 degrees for the bending type, folding inward to 90 degrees for the sharp folding type, each 180 degrees rotation for the kirigami type, and each press for the stack-up type.

The result shows that bending and sharp folding structures performed well with a resistive change of less than 10% after 100 uses, making them suitable for prototyping (Figure 7-e). However, kirigami and stack-up methods did not perform well, with kirigami showing a large resistance change of more than 200% at 40 uses and stack-up showing measurable open-circuit resistance (Figure 7-f). The unanticipated decrease in resistance when bending 80 to 100 times and when sharply folding 60 to 80 times may be attributed to turbulent connection resistance caused by the alligator clip. It is concluded that bending and sharp folding are safe for long-term usage, while kirigami and stack-up can only be used shortly with less than 40 times.

6 APPLICATION

6.1 Tangible User Interface for Rapid Prototype



Figure 8: Tangible user interface prototypes: (a)(b) game controllers; (c) an music oscillator.

In Figure 8, we demonstrated several tangible user interface designs through the occasion of our method using triple-layered 14B pencil traces and different primitive structures. The interactive elements include buttons, sliders, and knobs, and can be customized in different sizes, shapes, or colors. In Figure 8-a, four resistive buttons control the movement of a pacman in four directions. In Figure 8-b, an eight-key resistive controller simulates joystick operation, controlling the shooting direction of a plane. In Figure 8-c, the knob and sliders are used to create an oscillator. The knob, adopting the resistive kirigami type, sets the volume and the three-dimensional sliders set the frequency and wave type. The buttons, made of corrugated paper, produce a spring-like cushioning and lift when released, providing good tactile feedback.

6.2 Interactive Picture Book

The Pencil-E method is suitable for education as it is easy to obtain and combine general art and craft materials such as colored paper and pencils. Figure 9-a shows an interactive picture book initiated by covering a printed picture on a capacitive touchpad, which is



Figure 9: Use capacitive sensors to build (a) an interactive picture book; (b) an interactive painting.

made by double-layered 14B pencil traces on sulfate paper. When the user clicks on interactive areas, the computer displays corresponding information. It is highly flexible since the sensor is only one paper thin and can be attached to any printed surface. The capacitive sensor can also be built with free-form drawing (Figure 9-b), serving as both a functional base and artwork.

6.3 Water Level Gauge

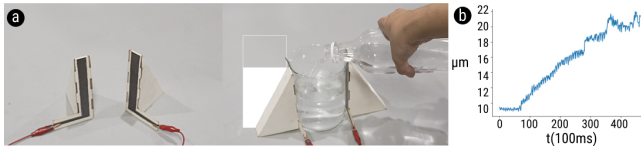


Figure 10: (a) Use a capacitive sensor to create a water level gauge; (b) the capacity outcome of the gauge.

As shown in Figure 10, our method can be applied to create measuring instruments with low precision, like a water gauge. The gauge is a capacitor that uses water and air as a dielectric medium. Its capacitance changes when water is added or removed. This type of gauge can be easily created and customized to fit the shape of containers.

6.4 Wearable Sensor

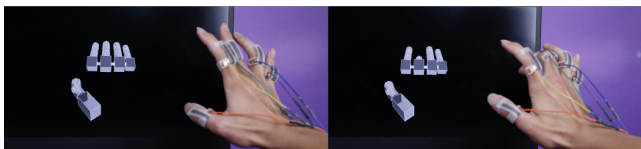


Figure 11: Capacitive sensors can be used as wearable sensors to detect gestures.

Figure 11 shows a set of ring-like wearable sensors made of sulfate paper and double-layered 14B pencil traces that detect the bending of the fingers. They are composed of a bridged paper structure and a pencil trace facing the finger joint. The sensor is a capacitor whose plate is the pencil layer and ground. When the finger bends, the pencil trace connects the finger joint, making the user's body a new capacitor plate, and increasing the capacity of the "sensor-ground" system significantly by enlarging the plate area. This method is low-cost, lightweight, easy to make, simple in structure, and with good sensing accuracy.

7 CONCLUSION

In this paper, we present Pencil-E, a low-cost, accessible, and expressive fabrication method that uses paper crafting techniques and pencil sketches to create customized capacitive and resistive sensors. We also introduced different structures such as single-layer, multi-layer, and constructive ones, and promised extensive interactions containing touch, slide, twitch, and press. Leveraging the use of pencil and paper, Pencil-E can introduce a broader community to personal fabrication and expand the range of methods and materials for rapid prototyping.

There are some limitations to our method, such as the lack of thorough evaluation of the durability and contact resistance of connectors like alligator clips and copper coil tapes. Additionally, it is hard for us to establish a precise relationship between continuous interaction and sensor output. We aim to improve the durability and precision of our method and develop output elements that can be used under slight currents, like LEDs and buzzers, to expand the possible range of scenarios.

ACKNOWLEDGMENTS

This project is supported by the National Natural Science Foundation of China (No. 62202423), and the Fundamental Research Funds for the Central Universities (No. 2022FZZX01-22).

REFERENCES

- [1] Zekun Chang, Tung D Ta, Koya Narumi, Heeju Kim, Fuminori Okuya, Dongchi Li, Kunihiro Kato, Jie Qi, Yoshinobu Miyamoto, Kazuya Saito, et al. 2020. Kirigami haptic swatches: Design methods for cut-and-fold haptic feedback mechanisms. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [2] Lucy E. Dunne, Kaila Bibeau, Lucie Mulligan, Ashton Frith, and Cory Simon. 2012. Multi-Layer e-Textile Circuits. In *Proceedings of the 2012 ACM Conference on Ubiquitous Computing (Pittsburgh, Pennsylvania) (UbiComp '12)*. Association for Computing Machinery, New York, NY, USA, 649–650.
- [3] Taesik Gong, Yeonsu Kim, Jinwoo Shin, and Sung-Ju Lee. 2019. Metasense: few-shot adaptation to untrained conditions in deep mobile sensing. In *Proceedings of the 17th Conference on Embedded Networked Sensor Systems*. 110–123.
- [4] Daniel Groeger and Jürgen Steimle. 2019. LASEC: Instant Fabrication of Stretchable Circuits Using a Laser Cutter. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [5] Freddie Hong, Connor Myant, and David E Boyle. 2021. Thermoformed Circuit Boards: Fabrication of highly conductive freeform 3D printed circuit boards with heat bending. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–10.
- [6] Cedric Honnet, Hannah Perner-Wilson, Marc Teyssier, Bruno Fruchard, Jürgen Steimle, Ana C Baptista, and Paul Strohmeier. 2020. PolySense: Augmenting Textiles with Electrical Functionality using In-Situ Polymerization. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [7] Sam Jacoby and Leah Buechley. 2013. Drawing the electric: storytelling with conductive ink. In *Proceedings of the 12th International Conference on Interaction Design and Children*. 265–268.
- [8] Tatsuya Kaihou and Akira Wakita. 2013. Electronic origami with the color-changing function. In *Proceedings of the second international workshop on Smart material interfaces: another step to a material future*. 7–12.
- [9] Ting-Kuo Kang. 2014. Tunable piezoresistive sensors based on pencil-on-paper. *Applied Physics Letters* 104, 7 (2014), 073117.
- [10] Hsin-Liu Kao, Christian Holz, Asta Roseway, Andres Calvo, and Chris Schmandt. 2016. DuoSkin: rapidly prototyping on-skin user interfaces using skin-friendly materials. In *Proceedings of the 2016 ACM International Symposium on Wearable Computers*. 16–23.
- [11] Nicholas A Knouf. 2017. Felted Paper Circuits Using Joomchi. In *Proceedings of the Eleventh International Conference on Tangible, Embedded, and Embodied Interaction*. 443–450.
- [12] Narendra Kurra and Giridhar U Kulkarni. 2013. Pencil-on-paper: electronic devices. *Lab on a Chip* 13, 15 (2013), 2866–2873.
- [13] Xinqin Liao, Qingliang Liao, Xiaoqin Yan, Qijie Liang, Haonan Si, Minghua Li, Hualin Wu, Shiyao Cao, and Yue Zhang. 2015. Flexible and highly sensitive strain

- sensors fabricated by pencil drawn for wearable monitor. *Advanced Functional Materials* 25, 16 (2015), 2395–2401.
- [14] Lijuan Liu, Jiahao Guo, Chao Zhang, Zhangzhi Wang, Pinqi Zhu, Tuo Fang, Junwu Wang, Cheng Yao, and Fangtian Ying. 2021. ElectroPaper: Design and Fabrication of Paper-Based Electronic Interfaces for the Water Environment. *Electronics* 10, 5 (2021), 604.
 - [15] Yiyue Luo, Kui Wu, Tomás Palacios, and Wojciech Matusik. 2021. KnitUI: Fabricating interactive and sensing textiles with machine knitting. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–12.
 - [16] Hyunjooh Oh, Tung D Ta, Ryo Suzuki, Mark D Gross, Yoshihiro Kawahara, and Lining Yao. 2018. PEP (3D Printed Electronic Papercrafts): An Integrated Approach for 3D Sculpting Paper-Based Electronic Devices. In *CHI*. 441.
 - [17] Simon Olberding, Sergio Soto Ortega, Klaus Hildebrandt, and Jürgen Steimle. 2015. Foldio: Digital fabrication of interactive and shape-changing objects with foldable printed electronics. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology*. 223–232.
 - [18] Narjes Pourjafarian, Marion Koelle, Bruno Fruchard, Sahar Mavali, Konstantin Klamka, Daniel Groeger, Paul Strohmeier, and Jürgen Steimle. 2021. Bodystylus: freehand on-body design and fabrication of epidermal interfaces. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–15.
 - [19] Jie Qi, Natalie Freed, Tiffany Tseng, Fay Shaw, Barbara Liedahl, Becca Rose Glowacki, and Yoshihiro Kawahara. 2021. Exquisite Circuits: Collaborative Electronics Design through Drawing Games. In *Creativity and Cognition*. 1–14.
 - [20] Jie Qi, Andrew “bunnie” Huang, and Joseph Paradiso. 2015. Crafting technology with circuit stickers. In *Proceedings of the 14th International Conference on Interaction Design and Children*. 438–441.
 - [21] Jonathan E Thompson. 2017. Pencil-on-paper capacitors for hand-drawn rc circuits and capacitive sensing. *Journal of Chemistry* 2017 (2017).
 - [22] Rao. R. Tummala. 2001. Fundamentals of Microsystems Packaging.
 - [23] Guanyun Wang, Tingyu Cheng, Youngwook Do, Humphrey Yang, Ye Tao, Jianzhe Gu, Byoungkwon An, and Lining Yao. 2018. Printed paper actuator: A low-cost reversible actuation and sensing method for shape changing interfaces. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–12.
 - [24] Guanyun Wang, Fang Qin, Haolin Liu, Ye Tao, Yang Zhang, Yongjie Jessica Zhang, and Lining Yao. 2020. MorphingCircuit: An integrated design, simulation, and fabrication workflow for self-morphing electronics. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 4 (2020), 1–26.
 - [25] YQ Wang, KH Low, HLJ Pang, KH Hoon, FX Che, and YS Yong. 2006. Modeling and simulation for a drop-impact analysis of multi-layered printed circuit boards. *MICROELECTRONICS RELIABILITY* 46, 2-4 (FEB-APR 2006), 558–573. <https://doi.org/10.1016/j.microrel.2005.05.007>
 - [26] Yanan Wang, Shijian Luo, Yujia Lu, Hebo Gong, Yexing Zhou, Shuai Liu, and Preben Hansen. 2017. AnimSkin: fabricating epidermis with interactive, functional and aesthetic color animation. In *Proceedings of the 2017 Conference on Designing Interactive Systems*. 397–401.
 - [27] Clement Zheng, HyunJoo Oh, Laura Devendorf, and Ellen Yi-Luen Do. 2019. Sensing kirigami. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. 921–934.